BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Practical Programming Project

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. For valid ages 13-18, give an ordinary normal test value.

2. Where can a local variable be used?

A) In every program
B) Within its local scope
C) Across every network
D) In all other subprograms

3. "Data interception is only copying a paper document from a filing cabinet." Correct this. Explain why.

4. A new user needs to open and sort files. "A user interface replaces all CPU instructions." Correct this. How does an interface help?

Today's objectives

Design, implement and test a small working program. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Design
Define success criteria, inputs, algorithms and expected outputs before writing code.

Implementation
Build a small working feature using appropriate constructs.

Testing
Record test inputs, expected outputs, actual outcomes and pass/fail results.

Refinement
Use test evidence to correct program faults and improve reliability.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Design	Define success criteria, inputs, algorithms and expected outputs before writing code. Consider this in a ticket-sales mini project.
Implementation	Build a small working feature using appropriate constructs. Consider this in a pupil quiz project.
Testing	Record test inputs, expected outputs, actual outcomes and pass/fail results. Consider this in a school lunch calculator.
Refinement	Use test evidence to correct program faults and improve reliability. Consider this in a sports points tracker.

I Do - One worked example

A ticket-sales mini project needs design and implementation. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1].

3. Implementation [1]; Build a small working feature using appropriate constructs. [1].

4. Give a specific point and a linked reason for each feature.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a pupil quiz project, explain the purpose of design. [2 marks]

2. A school lunch calculator needs implementation. Explain its role. [2 marks]

3. A sports points tracker uses testing and refinement. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Trace the OCR ERL program, test a new input and improve the code.

```
ticketPrice = 5
quantity = 3
total = ticketPrice * quantity
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how implementation and testing help in a book-return reminder. [4 marks]

Q4. Explain the roles of design, testing and refinement in a classroom club booking. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Planning a program involves identifying success criteria, inputs, algorithms and expected outputs before coding. ☐ T ☐ F
2. Define success criteria, inputs, algorithms and expected outputs before writing code ☐ T ☐ F
3. Build a small working feature using appropriate constructs ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 15 is a representative valid value away from the boundaries. ☐

Q2 B - Within its local scope ☐

Q3 Capturing data during transmission, potentially exposing confidential information. ☐

Q4 A user interface lets people use the computer, through GUI controls or typed commands. Other OS tasks manage devices, accounts, memory and files. ☐

Code First - Model

Original output: 15.
Updated tests: At 0: total 0; at 4: total 20. A negative value should print an error after validation is added.
Write the expected and actual output separately.

We Do - Check the model

1. Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1]. ☐

2. Implementation [1]; Build a small working feature using appropriate constructs. [1]. ☐

3. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Design [1]; Testing [1]. ☐

Q3 Implementation [1]; Build a small working feature using appropriate constructs. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. ☐

Q4 Design [1]; Define success criteria, inputs, algorithms and expected outputs before writing code. [1]. Testing [1]; Record test inputs, expected outputs, actual outcomes and pass/fail results. [1]. Refinement [1]; Use test evidence to correct program faults and improve reliability. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: